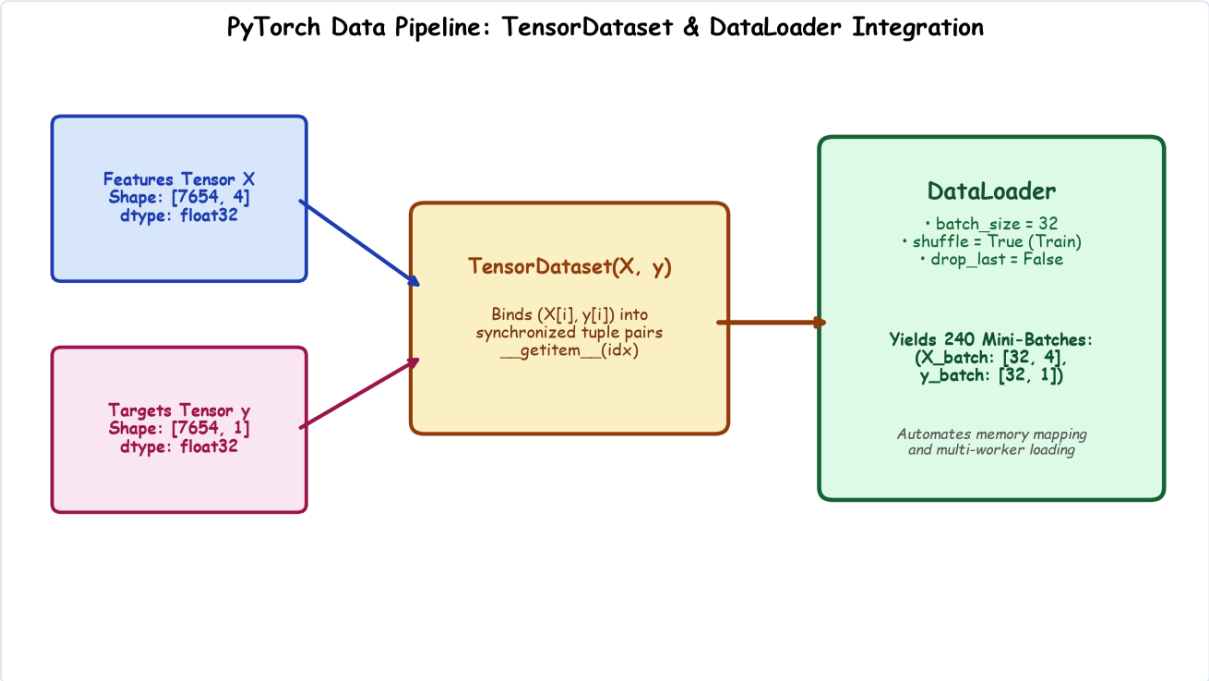
Submodule 38.03: PyTorch TensorDataset & DataLoader (Hinglish)

1. TensorDataset aur DataLoader kya hain?

Deep learning me saara data ek saath memory me nahi daalte. Data ko batches me todne ke liye PyTorch ke do tools use hote hain:

1. `TensorDataset` : Yeh features (X) aur labels (y) ko ek sath jod kar pairs (x_i, y_i) bana deta hai.
2. `DataLoader` : Yeh ek intelligent generator hai jo data ke 32-32 samples ke tukde (mini-batches) banata hai aur unhe shuffle karta hai.



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2. Implementation Code

```
from torch.utils.data import TensorDataset, DataLoader

# 1. Dataset create karna
train_dataset = TensorDataset(X_train_tensor, y_train_tensor)
test_dataset = TensorDataset(X_test_tensor, y_test_tensor)

# 2. DataLoader create karna (Batch size = 32)
train_loader = DataLoader(train_dataset, batch_size=32, shuffle=True)
```

```
test_loader = DataLoader(test_dataset, batch_size=32, shuffle=False)

print("Total Batches in 1 Epoch:", len(train_loader))  # 240 batches
```

- **`shuffle=True` kyun lagaya?**: Training ke time par data ko har epoch me mix (shuffle) karna zaroori hota hai taaki model kisi specific sequence ko yaad na kar le (overfitting se bache).
- Testing me `shuffle=False` rakha jata hai kyunki evaluation me order se koi farak nahi padta.\n